

- ① $\nabla(F \cdot G) = F \times (\nabla \times G) - (\nabla \times F) \times G + (G \cdot \nabla)F + (F \cdot \nabla)G$
- ② $\nabla \cdot (F \times G) = (\nabla \times F) \cdot G - F \cdot (\nabla \times G)$
- ③ $\text{Curl} \rightarrow \nabla \times F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \end{vmatrix} \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} = \left[\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right] i + \left[\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} \right] j + \left[\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} \right] k$
- ④ $\nabla \times (bF) = (\nabla b) \times F + b(\nabla \times F)$
- ⑤ $\nabla \times (F \times G) = F(\nabla \cdot G) - (\nabla \cdot F)G + (G \cdot \nabla)F - (F \cdot \nabla)G$
- ⑥ Laplacian $\rightarrow \nabla^2 b = \nabla \cdot (\nabla b) = \left(\frac{\partial^2 b}{\partial x^2} + \frac{\partial^2 b}{\partial y^2} + \frac{\partial^2 b}{\partial z^2} \right)$
 $\nabla^2 (bg) = b \nabla^2 g + 2 \nabla b \cdot \nabla g + g \nabla^2 b$
- ⑦ $\nabla \cdot (\nabla \times b) = 0$ div. of curl
 $\nabla \times (\nabla b) = 0$ curl of grad. deg
- $\nabla \cdot (b(\nabla g \times \nabla h)) = \nabla b \cdot (\nabla g \times \nabla h)$
- $\nabla \cdot (b \nabla g - g \nabla b) = b \nabla^2 g - g \nabla^2 b$
- $\nabla \times (\nabla \times F) = \nabla(\nabla \cdot F) - \nabla^2 F$
- ⑧ Gauss's theorem (divergence theorem)
 $\iint_S F \cdot dS = \iiint_V (\nabla \cdot F) dV$
- ⑨ Gauss's law $\rightarrow \nabla \cdot E = \rho / \epsilon_0$
- ⑩ Stokes's theorem $\rightarrow \oint_C F \cdot dl = \iint_S (\nabla \times F) \cdot dS$
- ⑪ FL $\rightarrow \nabla \times E = -\frac{\partial B}{\partial t}$
- ⑫ AML $\rightarrow \nabla \times H = J + \frac{\partial D}{\partial t}$
- ⑬ $\oint_S E \cdot da = Q_{enc} / \epsilon_0$ } G.L
 $\nabla \cdot E = \rho / \epsilon_0$
 $\oint_C E \cdot dl = 0$
 $\nabla \times E = 0$
- ⑭ $\nabla^2 V = -\rho / \epsilon_0$ } $\rightarrow$ POISSON'S E.Q.
 $\nabla^2 A = -\mu_0 J$
 $\nabla^2 V = 0 \rightarrow$ Laplace's eqn

$$\textcircled{13} \quad W = Q \Delta V \quad \nabla^2 = -\rho/\epsilon_0 \quad \nabla \cdot E = \rho/\epsilon_0$$

$$V = K \int \frac{1}{r} d\tau = \frac{KQ}{r} \quad E = K \frac{Q}{r^2} \quad \nabla \times E = 0$$

$$\textcircled{14} \quad \text{Dipole moment} \rightarrow P = \alpha E = q \times d$$

↳ atomic polarisability

$$\textcircled{17} \quad \text{Torque} \rightarrow \tau = qd \times E = P \times E$$

$$\textcircled{15} \quad \nabla \cdot E = \frac{\rho}{\epsilon_0} = \frac{\rho_{\text{free}} + \rho_{\text{bound}}}{\epsilon_0} \quad (\rho_{\text{bound}} = -\nabla \cdot P)$$

$$\epsilon_0 \nabla \cdot E = -\nabla \cdot P + \rho_{\text{free}}$$

$$\nabla \cdot (\epsilon_0 E + P) = \rho_{\text{free}}$$

$$\nabla \cdot D = \rho_{\text{free}}$$

$D = \epsilon_0 E + P \rightarrow$ Electric Displacement Vector

$$\textcircled{19} \quad \nabla \times D = \epsilon_0 (\nabla \times E) + \nabla \times P = \nabla \times P$$

$$\textcircled{20} \quad \chi_e = \frac{\alpha}{\epsilon_0} \rightarrow \text{electric susceptibility}$$

$$P = \alpha E = \chi_e \epsilon_0 E$$

$$\therefore D = \epsilon_0 E + P = \epsilon_0 E + \chi_e \epsilon_0 E = \epsilon_0 E (1 + \chi_e) = \epsilon E$$

$$\boxed{\epsilon = \epsilon_0 (1 + \chi_e)}$$

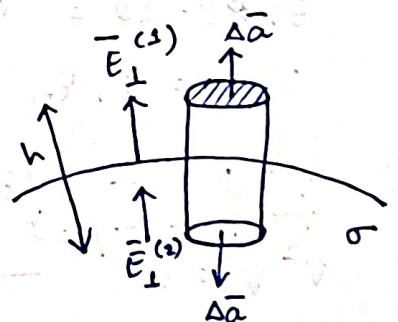
$$\epsilon_r = 1 + \chi_e = \epsilon / \epsilon_0$$

$\textcircled{21}$ BOUNDARY CONDITION :

$\textcircled{A}$ $E \rightarrow$

(a) Normal component

$$\oint_s E \cdot da = \oint_{\text{upper}} E \cdot da + \oint_{\text{lower}} E \cdot da + \oint_{\text{sides}} E \cdot da$$



$$\oint_C \mathbf{E} \cdot d\mathbf{a} = \oint_{\text{upper}} \mathbf{E} \cdot d\mathbf{a} + \oint_{\text{lower}} \mathbf{E} \cdot d\mathbf{a} = \oint_C \frac{\sigma d\mathbf{a}}{\epsilon_0}$$

$$E_{\perp}^{(1)} da + E_{\perp}^{(2)} (-da) = \frac{\sigma da}{\epsilon_0}$$

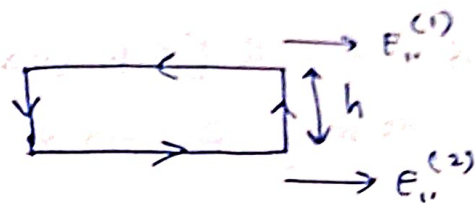
$$E_{\perp}^{(1)} - E_{\perp}^{(2)} = \frac{\sigma}{\epsilon_0} \therefore \text{discontinuous}$$

(b) Tangential Component

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = 0$$

$$E_{\parallel}^{(1)}(-l) + E_{\parallel}^{(2)}(l) = 0$$

$$E_{\parallel}^{(1)} = E_{\parallel}^{(2)} \therefore \text{continuous always}$$



⑧ $\mathbf{D} \rightarrow$

(a) Normal Component

$$\oint_S \mathbf{D} \cdot d\mathbf{a} = Q_{\text{free, enc}} = \oint_S \sigma_b \cdot d\mathbf{a}$$

$$D_{\perp}^{(1)} da + D_{\perp}^{(2)} (-da) = \sigma_b \cdot da$$

$$D_{\perp}^{(1)} - D_{\perp}^{(2)} = \sigma_b \neq 0 \therefore \text{discontinuous}$$

(b) Tangential Component

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$$

$$\oint_C \mathbf{D} \cdot d\mathbf{l} = \epsilon_0 \oint_C \mathbf{E} \cdot d\mathbf{l} + \oint_C \mathbf{P} \cdot d\mathbf{l}$$

$$\oint_C \mathbf{D} \cdot d\mathbf{l} = \oint_C \mathbf{P} \cdot d\mathbf{l}$$

$$\oint_C (\mathbf{D} - \mathbf{P}) \cdot d\mathbf{l} = 0$$

$$\oint_C \mathbf{C} \cdot d\mathbf{l} = 0$$

$$\boxed{\mathbf{C} = \mathbf{D} - \mathbf{P}}$$

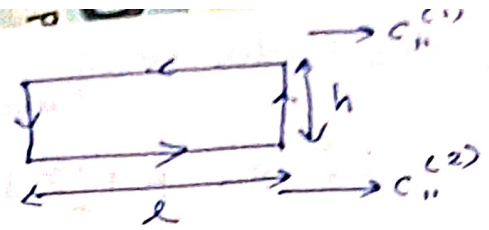
only longer sides contribute.

$$C_{||}^{(1)}(-l) + C_{||}^{(2)}(l) = 0$$

$$\Rightarrow C_{||}^{(1)} = C_{||}^{(2)}$$

$$\therefore D_{||}^{(1)} - P_{||}^{(1)} = D_{||}^{(2)} - P_{||}^{(2)}$$

$\therefore$ discontinuous unless $P_{||}^{(1)} = P_{||}^{(2)}$



(22) $F = Q [E + (v \times B)] \rightarrow$ LORENTZ FORCE LAW

mag. $F \rightarrow$ work

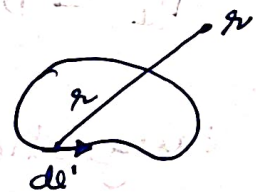
$$\begin{aligned} dW_{\text{mag}} &= f_{\text{mag}} \cdot dl \\ &= Q (v \times B) \cdot v dt = 0 \checkmark \\ &[\because (v \times B) \text{ is } \perp \text{ to } v] \end{aligned}$$

F_{mag} on a segment of current-carrying wire —

$$\begin{aligned} F_{\text{mag}} &= \int dq (v \times B) = \int I dt (v \times B) \\ &= \int I (dl \times B) = I (l \times B) \end{aligned}$$

(23) BIOT-SAVART LAW

$$B(r) = \frac{\mu_0}{4\pi} \int I \frac{dl' \times \hat{r}}{r^2} \rightarrow \text{LINE CURRENT}$$



$$K = \text{Surface current density} = \frac{dI}{dl_{\perp}}$$

$$K = \sigma v \quad \left(\begin{array}{l} \sigma \rightarrow \text{surface charge density} \\ v \rightarrow \text{velocity} \end{array} \right)$$

$$B(r) = \frac{\mu_0}{4\pi} \int \frac{K(r') \times \hat{r}}{r^2} da' \rightarrow \text{SURFACE CURRENT}$$

• $J = \frac{dI}{da_L} \rightarrow$ volume current density

• $B(r) = \frac{\mu_0}{4\pi} \int \frac{J(r') \times \hat{r}}{r'^2} dv' \rightarrow$ VOLUME CURRENT

• $\oint_S J \cdot da = \int_V (\nabla \cdot J) dv$

(24) CONTINUITY EQUATION -

$\int_V (\nabla \cdot J) dv = -\frac{d}{dt} \int_V \rho \cdot dv$



$= -\int_V \left(\frac{\partial \rho}{\partial t} \right) dv$

$\therefore \nabla \cdot J = -\frac{\partial \rho}{\partial t}$

OR

$Q = \int_V \rho \cdot dv$

$I = \oint_S J \cdot dA$

$\frac{dQ}{dt} = -I$

$\frac{dQ}{dt} = \frac{d}{dt} \int_V \rho \cdot dv = \int_V \frac{\partial \rho}{\partial t} \cdot dv = -\oint_S J \cdot dA$

$\int_V \frac{\partial \rho}{\partial t} \cdot dv = -\int_V (\nabla \cdot J) dv$

$\therefore -\frac{\partial \rho}{\partial t} = \nabla \cdot J$

(25) CURL OF B -

$\oint_S (\nabla \times B) \cdot da = \oint_C B \cdot dl$

(circular path of radius r)



$= \frac{\mu_0 I}{2\pi r} \oint_C dl = \frac{\mu_0 I}{2\pi r} \cdot 2\pi r = \mu_0 I \checkmark$

$\oint_C B \cdot dl = \mu_0 I_{enc}$

Also, $dl_{||} = r d\phi$

$$\oint_C \mathbf{B} \cdot d\mathbf{l} = \oint_C B r d\phi$$

$$= \oint_C \frac{\mu_0 I}{2\pi r} r d\phi$$

$$= \frac{\mu_0 I_{enc}}{2\pi} \oint_C d\phi = \frac{\mu_0 I_{enc}}{2\pi} (2\pi)$$

$$\boxed{\oint_C \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{enc}}$$

→ Ampere's Law

ACL

~~16~~ • AL

① electrodynamics
(time-varying field/I)

② I d✓

③ I J & $\frac{\partial E}{\partial t} \rightarrow$ changing flux = I

④ q conservedⁿ
(continuity eqⁿ ✓)

⑤ one of the 4 M eqⁿ

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$

$$\oint_C \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_c + \mu_0 \epsilon_0 \frac{d\phi_c}{dt}$$

① magnetostatics
(steady I)

② X

③ J ✓ $\rho = \rho$

④ inconsistent with
q consⁿ

⑤ X = $\frac{\rho - \rho}{\epsilon_0}$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$$

$$\oint_C \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{enc}$$

$$\oint_S (\nabla \times B) \cdot d\mathbf{a} = \oint_C B \cdot d\mathbf{l} = \mu_0 I_{enc} = \mu_0 \int_S J \cdot d\mathbf{a}$$

$$\oint_S (\nabla \times B) \cdot d\mathbf{a} = \mu_0 \int_S J \cdot d\mathbf{a}$$

$$\boxed{\nabla \times B = \mu_0 J} \quad \text{Ampere}$$

$$\oint_S B \cdot d\mathbf{a} = 0 \quad \checkmark$$

$$\nabla \cdot B = 0 \quad \checkmark$$

Curl of B

$$\nabla \times B = \mu_0 J$$

$$\oint_C B \cdot d\mathbf{l} = \mu_0 I_{enc}$$

Div. of B

$$\nabla \cdot B = 0 \rightarrow \text{GL for magnetism}$$

$$\oint_S B \cdot d\mathbf{a} = 0$$

• Derⁿ of $\nabla \cdot B = 0$ $\rightarrow$

As $\vec{B}$ lines always form closed loops.

$\therefore$ net ϕ_B passing out $= 0$.

$$\phi_B = \iint_S B \cdot d\mathbf{s} = 0$$

Now, Applying Divergence Theorem $\rightarrow$

$$\iint_S B \cdot d\mathbf{s} = \iiint_V (\nabla \cdot B) dV = 0 \quad \checkmark$$

$$\therefore \nabla \cdot B = 0$$

$\rightarrow$ ① Magnetic monopoles $\times$ exist

② Closed flux lines:

OR

$$\nabla \cdot B = \frac{\mu_0}{4\pi} \int \frac{J(\mathbf{r}') \times \hat{r}}{r^2} dv'$$

$$\nabla \cdot B = \frac{\mu_0}{4\pi} \int \nabla \cdot \left(J \times \frac{\hat{r}}{r^2} \right) dv'$$

$$= \frac{\mu_0}{4\pi} \int \left[\frac{\hat{r}}{r^2} \cdot (\nabla \times J) - J \cdot \left(\nabla \times \frac{\hat{r}}{r^2} \right) \right] dv'$$

$$= \frac{\mu_0}{4\pi} \int \left[J \cdot \left(\nabla \times \frac{\hat{r}}{r^2} \right) \right] dv'$$

$$\nabla \times \frac{\hat{r}}{r^2} = \frac{1}{r^2} (\nabla \times \hat{r}) + \nabla \left(\frac{1}{r^2} \right) \times \hat{r} = -\frac{2}{r^3} (\hat{r} \times \hat{r}) = 0$$

$$\therefore \nabla \cdot B = 0$$

#

$$\nabla \cdot E = \rho / \epsilon_0 \rightarrow GL$$

$$\nabla \cdot B = 0$$

$$\nabla \times E = 0$$

$$\nabla \times B = \mu_0 J \rightarrow AL$$

4 Maxwell eqⁿs.

(16)

VECTOR POTENTIAL —

$$F = q_0 E$$

$$dW = F \cdot dl = q_0 E \cdot dl$$

$$dV = -\frac{dW}{q_0} = -\frac{q_0 E \cdot dl}{q_0} = -E \cdot dl$$

$$\text{Now, } dV = \nabla V \cdot dl$$

$$\therefore \nabla V \cdot dl = -E \cdot dl$$

$$\therefore \boxed{E = -\nabla V}$$

$$(27) \quad A = \frac{\mu_0}{4\pi} \int \frac{J}{r} dV$$

$$B = \nabla \times A$$

$$\nabla \cdot A = 0$$

(28) BOUNDARY CONDITION FOR B -

(a) Normal component

$$\nabla \cdot B = 0$$

$$\oint_S B \cdot da = 0$$

$$\oint_S B \cdot da = \oint_{\text{upper}} B \cdot da + \oint_{\text{lower}} B \cdot da$$

$$B_{\perp}^{(1)} \Delta a + B_{\perp}^{(2)} (-\Delta a) = 0$$

$$B_{\perp}^{(1)} = B_{\perp}^{(2)} \quad \therefore \text{continuous}$$

(b) Tangential component

$$\oint_C B \cdot dl = \mu_0 I_{\text{enc}}$$

$$B_{\parallel}^{(1)} (-l) + B_{\parallel}^{(2)} (l) = -\mu_0 (kl) \quad \left[\int k \cdot dl = I \right]$$

$$B_{\parallel}^{(1)} - B_{\parallel}^{(2)} = \mu_0 k$$

$\therefore$ discontinuous

$$(29) \quad N = 2\sigma \times F$$

$$= 2 \left(\frac{a}{2} \right) f \sin \theta \hat{x} = af \sin \theta \hat{x}$$

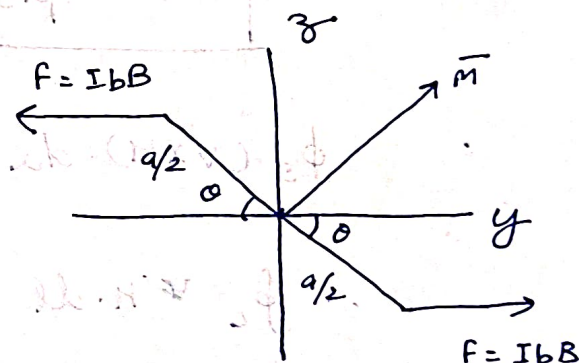
$$\therefore f = IbB$$

$$\therefore N = a(IbB) \sin \theta \hat{x}$$

$$= I(ab) B \sin \theta \hat{x}$$

$$= I \cdot A \cdot B \cdot \sin \theta \hat{x}$$

$$(ab = A)$$



$m = IA \rightarrow$ magnetic dipole moment

$$\therefore N = mB \sin \theta \hat{n} = \vec{m} \times \vec{B}$$

$$N = p \times E$$

(30) $M \rightarrow$ magnetic dipole moment per unit volume
 $\downarrow$
 magnetization

$$B = \nabla \times A \quad (\text{like } E = -\nabla V)$$

$\hookrightarrow A$ is the vector potential

$$J_b = \nabla \times M$$

$$K_b = M \times \hat{n}$$

$$\nabla \times B = \mu_0 J = \mu_0 (J_b + J_f)$$

$$\frac{1}{\mu_0} (\nabla \times B) = \nabla \times M + J_f$$

$$\nabla \times \left(\frac{B}{\mu_0} - M \right) = J_f$$

$$\nabla \times H = J_f$$

$$H = \frac{B}{\mu_0} - M$$

$\rightarrow$ magnetic field strength (A/m)

$$\oint_S (\nabla \times H) \cdot d\mathbf{a} = \oint_S J_f \cdot d\mathbf{a} = I_{f, \text{enc}}$$

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = I_{f, \text{enc}}$$

$$\nabla \cdot H = \nabla \cdot \left(\frac{B}{\mu_0} - M \right) = -\nabla \cdot M$$

(31) BOUNDARY CONDITION -

$$\nabla \cdot (H + M) = 0 \quad \text{Normal}$$

$$H_{\perp}^{(1)} - H_{\perp}^{(2)} = M_{\perp}^{(2)} - M_{\perp}^{(1)} \rightarrow \text{discontinuous}$$

$$\oint_C H \cdot dl = I_{b, \text{enc}} \quad \text{Tangential}$$

$$H_{\parallel}^{(1)}(-l) + H_{\parallel}^{(2)}(l) = -K_b l$$

$$H_{\parallel}^{(1)} - H_{\parallel}^{(2)} = K_b \rightarrow \text{discontinuous}$$

$$\# \quad H = \left(\frac{B}{\mu_0} - M \right)$$

$$\nabla \cdot H = -\nabla \cdot M$$

$$\nabla \times H = J_b$$

$$\nabla \times M = J_b$$

$$\# \quad \nabla \cdot E = \rho / \epsilon_0$$

$$\nabla \cdot D = \rho_b$$

$$\nabla \times E = 0$$

$$\nabla \times D = \nabla \times P$$

$$\nabla \cdot B = 0$$

$$\nabla \cdot H = -\nabla \cdot M$$

$$\nabla \times B = \mu_0 J$$

$$\nabla \times H = J_b$$

$$E_{\perp}^{(1)} - E_{\perp}^{(2)} = \sigma / \epsilon_0$$

$$D_{\perp}^{(1)} - D_{\perp}^{(2)} = \sigma_b$$

$$E_{\parallel}^{(1)} = E_{\parallel}^{(2)}$$

$$D_{\parallel}^{(1)} - D_{\parallel}^{(2)} = P_{\parallel}^{(1)} - P_{\parallel}^{(2)}$$

$$B_{\perp}^{(1)} - B_{\perp}^{(2)} = 0$$

$$H_{\perp}^{(1)} - H_{\perp}^{(2)} = M_{\perp}^{(2)} - M_{\perp}^{(1)}$$

$$B_{\parallel}^{(1)} - B_{\parallel}^{(2)} = \mu_0 K_b$$

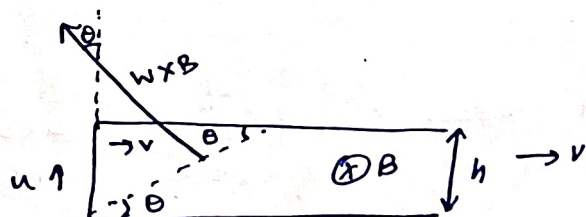
$$H_{\parallel}^{(1)} - H_{\parallel}^{(2)} = K_b$$

(32)

$$V_{\text{emf}} = \oint F \cdot dl$$

$$= \oint (w \times B) \cdot dl$$

$(u+v) \rightarrow \text{velocity}$



$$\boxed{\oint_{\text{emf}} = \oint B h} = - \frac{d\phi_B}{dt} = \oint E \cdot dl$$

$$\oint E \cdot dl = - \frac{d}{dt} \int B \cdot da = - \int \frac{\partial B}{\partial t} \cdot da$$

$$\int (\nabla \times E) \cdot da = - \int \frac{\partial B}{\partial t} \cdot da$$

$$\therefore \nabla \times E = - \frac{\partial B}{\partial t}$$

(33)

$$\nabla \cdot E = \frac{\rho}{\epsilon_0} \rightarrow \rho = \epsilon_0 (\nabla \cdot E)$$

$$\nabla \cdot J = - \frac{\partial \rho}{\partial t} = - \frac{\partial}{\partial t} (\epsilon_0 \nabla \cdot E) = - \nabla \cdot \left(\epsilon_0 \frac{\partial E}{\partial t} \right)$$

$$\nabla \cdot \left(J + \epsilon_0 \frac{\partial E}{\partial t} \right) = 0$$

$$\nabla \cdot J_c = 0$$

$$\boxed{J_d = \epsilon_0 \frac{\partial E}{\partial t}}$$

$$\therefore \nabla \times B = \mu_0 J + \mu_0 \epsilon_0 \frac{\partial E}{\partial t}$$

$$= \mu_0 J + \mu_0 J_d$$

$$= \mu_0 (J + J_d)$$

$$\hookrightarrow \text{displacement current} = \epsilon_0 \frac{\partial E}{\partial t}$$

Div of ∇ of any scalar field = 0
 # Div of ∇ of any vector = 0